

Semi-Lagrangian off-lattice Boltzmann method for weakly compressible flowsAndreas Krämer,^{1,*} Knut Küllmer,^{1,†} Dirk Reith,^{1,‡} Wolfgang Joppich,^{1,§} and Holger Foysi^{2,||}¹*Institute for Technology, Renewables and Energy-efficient Engineering (TREE), Bonn-Rhein-Sieg University of Applied Sciences, Grantham-Allee 20, 53757 Sankt Augustin, Germany*²*Department of Mechanical Engineering, University of Siegen, Paul-Bonatz-Straße 9-11, 57076 Siegen-Weidenau, Germany*

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The lattice Boltzmann method is a simulation technique in computational fluid dynamics. In its standard formulation, it is restricted to regular computation grids, second-order spatial accuracy, and a unity Courant-Friedrichs-Lewy (CFL) number. This paper advances the standard lattice Boltzmann method by introducing a semi-Lagrangian streaming step. The proposed method allows significantly larger time steps, unstructured grids, and higher-order accurate representations of the solution to be used. The appealing properties of the approach are demonstrated in simulations of a two-dimensional Taylor-Green vortex, doubly periodic shear layers, and a three-dimensional Taylor-Green vortex.

DOI: [10.1103/PhysRevE.95.023305](https://doi.org/10.1103/PhysRevE.95.023305)**I. INTRODUCTION**

The lattice Boltzmann method (LBM) [1,2] is a computationally efficient approach to simulate fluid dynamics. It models the streaming and collision of fluid particles based on simplified mesoscopic dynamics. The kinetic description is tailored to solve the governing equations of fluid motion on the macroscopic scale, which enables the method to simulate general fluid dynamics in a simple and efficient manner. Becoming an increasingly versatile tool for fluid simulation, the LBM is successfully being applied in various areas of application, such as blood flow [3], molecular dynamics [4], multiphase flow [5], and fluid turbulence [6,7]. The ongoing work with the LBM includes large-scale simulations of fluid turbulence, which has been challenging classical simulation methods for decades [8,9]. However, the standard version of the LBM is restricted to regular grids, second-order accuracy in space and time, and a unity Courant-Friedrichs-Lewy (CFL) number.

Attempts to overcome these limitations have led to the development of so-called off-lattice Boltzmann methods (OLBMs). Most of the OLBMs solve the discrete Boltzmann equation by using finite-difference [10,11], finite-volume [12–15], finite-element [16,17], or discontinuous Galerkin schemes [18,19]. In doing so, they extend the LBM toward unstructured grids, higher-order spatial discretizations, and nonunity CFL numbers. Although these methods have improved over the past decade, they were usually only validated for simple two-dimensional flows and rarely evaluated in terms of computational efficiency.

In a preliminary study [20], we simulated a turbulent channel flow at $Re_\tau = 180$ using a high-order convergent spectral element discontinuous Galerkin (SEDG) LBM [19]

code. Our implementation of this relatively new OLBM [21] utilizes the high-performance library Trilinos [22] for parallelization. The simulations performed in this study spent more than 90% of the runtime on the SEDG streaming step, specifically on calculating fluxes and spatial derivatives. Moreover, the use of an explicit time integrator enforced small time steps, which resulted in an unacceptably long computation time.

Moving forward from that study requires understanding four problems that led to the high computational cost. First, the advective part of the discrete Boltzmann equation has to be solved for each of the discrete particle velocities, amounting to at least 15 partial differential equations in three dimensions. Second, the resulting high-dimensional spatial operator is evaluated repeatedly per time step due to the explicit Runge-Kutta time integration. Third, the explicit integrator enforces small time steps, as its critical CFL number is dictated by both the smallest grid spacing and the order of spatial discretization. Finally, the CFL number is formulated in terms of the particle velocities that are at least one order of magnitude larger than the flow velocity. These aspects add up to a costly spatial operator that has to be evaluated very often in a simulation.

These drawbacks apply to all previous high-order OLBMs. Furthermore, they may well be transferable to most of the so-called Eulerian OLBMs [23], i.e., those that explicitly discretize the spatial and temporal derivative in the discrete Boltzmann equation. According to this definition, Eulerian OLBMs include most of the previously published finite-difference, finite-volume, finite-element, and discontinuous Galerkin LBMs.

The above-mentioned issues can be resolved when switching from an Eulerian to a semi-Lagrangian framework. The semi-Lagrangian approach starts from a characteristic equation, such as the lattice Boltzmann equation, and obtains the solution at the departure points of the characteristic lines by interpolation; cf. Fig. 1. As the characteristic lines are linear in the LBM, the departure points are trivially found. The semi-Lagrangian method has two major advantages. It remains stable at large CFL numbers [24] and requires only one evaluation of the spatial operator per time step, i.e., the interpolation. These benefits make the method a convenient

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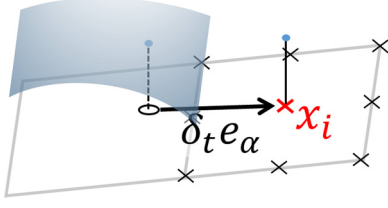


FIG. 1. Propagation of distribution function f_α in the semi-Lagrangian streaming. The streamed distribution function $f_\alpha(x_i, t)$ is reconstructed by a finite-element approximation at the departure point $(x_i - \delta_t e_\alpha, t - \delta_t)$. The finite-element approximation is depicted by the blue surface on the left cell. The crosses represent the support points on the right cell, here for $N = 2$.

tool for advection-dominated problems [24] and a promising candidate for enhancing the (inherently Lagrangian) LBM naturally and efficiently.

Two previous OLBMs already follow the same spirit under a different terminology [25,26]. They are called “interpolation-based” lattice Boltzmann methods, and their potential and efficiency were already demonstrated for two-dimensional flows that were simulated on unstructured grids [26]. The previous works use low-order accurate Taylor series-based interpolations to reconstruct the solution.

In this paper, we improve on the previous methods and present a semi-Lagrangian off-lattice Boltzmann method (SLLBM) that is based on finite-element reconstruction of the distribution functions. The proposed method solves the streaming step in a semi-Lagrangian framework, while the collision step remains local, as in both the standard LBM and some previous OLBMs [19,27]. The distributions at the departure points $f_\alpha(x - \delta_t e_\alpha, t - \delta_t)$ are reconstructed by Lagrangian finite elements, decoupling the size of the time step from the grid and allowing for high-order accurate solutions. In

the semi-Lagrangian finite-element formulation, the streaming step requires only one evaluation of the spatial operator per time step, and, in contrast to all explicit Eulerian LBMs, it does not pose a CFL constraint. This combination of high-order spatial accuracy and stability at large CFL numbers is unique in the LBM literature.

In Sec. II, the proposed method is formulated in two and three dimensions, followed by a study of its stability. Section III presents results for three benchmarks. First, the numerical errors in the SLLBM are evaluated for the case of a two-dimensional Taylor-Green vortex. Second, doubly periodic shear layers are studied to highlight the benefits of high-order spatial accuracy. Finally, the transition to turbulence in a three-dimensional Taylor-Green vortex is studied to validate the method in three dimensions. The paper focuses on demonstrating the flexibility of the SLLBM with respect to the spatial order and the time step, leaving complex bounded flows on irregular grids for a follow-up publication. One simulation is performed on an irregular grid to demonstrate that the method is applicable to more general grids. The results are discussed in Sec. IV. The paper ends with a conclusion (Sec. V).

II. METHODOLOGY

A. Discrete Boltzmann equation

The discrete Boltzmann equation with the Bhatnagar-Gross-Krook model [28] reads

$$\frac{\partial f_\alpha}{\partial t} + e_\alpha \cdot \nabla_x f_\alpha = -\frac{1}{\lambda} (f_\alpha - f_\alpha^{\text{eq}}), \quad (1)$$

where λ , f_α , and f_α^{eq} are the (dimensional) relaxation time, the particle distribution that is streaming with velocity e_α , and the local equilibrium distribution, respectively. In this work, we use particle velocities that are scaled by the parameter $\xi > 0$, so that

$$(e_0 \ e_1 \ \dots \ e_8) = \xi \begin{pmatrix} 0 & 1 & 0 & -1 & 0 & 1 & -1 & -1 & 1 \\ 0 & 0 & 1 & 0 & -1 & 1 & 1 & -1 & -1 \end{pmatrix}$$

in the D2Q9 model and

$$(e_0 \ e_1 \ \dots \ e_{18}) = \xi \begin{pmatrix} 0 & 1 & 0 & -1 & 0 & 0 & 0 & 1 & -1 & -1 & 1 & 1 & 1 & -1 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & 1 & 0 & 0 & 0 & 0 & -1 & 1 & 1 & -1 & -1 & 1 & 1 & -1 \\ 0 & 0 & 1 & 0 & -1 & 0 & 0 & 1 & 1 & -1 & -1 & 0 & 0 & 0 & 0 & 1 & 1 & -1 & -1 \end{pmatrix}$$

in the D3Q19 model. The equilibrium is calculated from the macroscopic moments

$$\rho = \sum_{\alpha=0}^{Q-1} f_\alpha \quad \text{and} \quad \rho u = \sum_{\alpha=0}^{Q-1} f_\alpha e_\alpha$$

as

$$f_\alpha^{\text{eq}} = w_\alpha \rho \left(1 + \frac{u e_\alpha}{c_s^2} - \frac{u^2}{2c_s^2} + \frac{(u e_\alpha)^2}{2c_s^4} \right),$$

with the speed of sound $c_s = \xi/\sqrt{3}$ and the weights $w_0 = 4/9$, $w_{1,\dots,4} = 1/9$, $w_{5,\dots,8} = 1/36$ in D2Q9 and $w_0 = 1/3$, $w_{1,\dots,6} = 1/18$, $w_{7,\dots,18} = 1/36$ in D3Q19. A Chapman-Enskog analysis of Eq. (1) shows that the macroscopic

moments approximately solve the Navier-Stokes equation with the kinematic viscosity $\nu = \lambda c_s^2$. The approximation is of order $O(\text{Ma}^2)$.

B. Lattice Boltzmann method

By integrating the discrete Boltzmann equation along characteristics with the trapezoidal rule and transforming the particle distribution functions as in [27], we arrive at the well-known lattice-BGK equation [29]

$$f_\alpha(x + \delta_t e_\alpha, t + \delta_t) = f_\alpha(x, t) - \frac{1}{\tau} [f_\alpha(x, t) - f_\alpha^{\text{eq}}(x, t)], \quad (2)$$

where $\tau := \lambda/\delta_t + 0.5$ is a dimensionless relaxation time. Note that this integration incorporates a second-order temporal error.

To simplify the numerical procedure, the lattice-BGK equation is split into the following:

(i) Collision:

$$\bar{f}_\alpha(x, t) = f_\alpha(x, t) - \frac{1}{\tau} [f_\alpha(x, t) - f_\alpha^{\text{eq}}(x, t)].$$

(ii) Streaming:

$$f_\alpha(x, t) = \bar{f}_\alpha(x - \delta_t e_\alpha, t - \delta_t).$$

The standard LBM uses a lattice spacing $\delta_x = \delta_t \xi$. Consequently, $x - \delta_t e_\alpha$ points from the lattice node x to one of its neighbors, which turns the streaming step into an extremely efficient index shift. In doing this, the discrete vector of degrees of freedom represents the point values of the distribution functions. This pointwise representation involves a spatial approximation error of order $O(\delta_x^2)$.

C. Semi-Lagrangian streaming step

The SLLBM inherits the splitting into collision and streaming steps from the standard LBM. In the spirit of the standard LBM, the BGK collision used in this work could be replaced by a more sophisticated model, such as the entropic model [30] or a multiple-relaxation-time model [31]. In contrast to the standard LBM, the present work performs the streaming step by a semi-Lagrangian advection, which comprises the standard streaming step on a regular grid as a special case.

1. Semi-Lagrangian method

The semi-Lagrangian method was originally developed for advection-dominated problems, where following the characteristics is more efficient than using time-marching schemes. This is particularly true for the lattice Boltzmann streaming step, where the characteristics are straight lines and the departure points are located at $x - \delta_t e_\alpha$.

The SLLBM represents the distribution functions by a finite element instead of a point-wise approximation. This has two major advantages: First, it decouples the computational grid from the time step, allowing unstructured grids as well as nonunity CFL numbers to be used; and second, it offers high-order accurate representations of the distribution functions, in contrast to previous interpolation-supplemented OLBM.

2. Definition of computational grid and finite elements

The present method works on quadrilateral and hexagonal computation grids, such as those used in finite-element simulations. To define the grid, the unit cell $\tilde{D} := [0, 1]^{\text{dim}}$ is mapped onto an arbitrary grid cell D_c through a bijective, multilinear mapping function $M_c : D_c \rightarrow \tilde{D}$. Each mapping function is associated with a single grid cell and defines its deformation, size, and position.

Support points $\tilde{x}_i$, $i = 1, \dots, (N+1)^{\text{dim}}$, are placed on the unit cell to define the finite elements. Concretely, this work uses Gauß-Lobatto-Legendre points [32]. We define the set of N th-order Lagrange polynomials $\tilde{\psi}_i$ on these support points,

so that $\tilde{\psi}_i(x_j) = \delta_{ij}$, where δ_{ij} is the Kronecker delta. In this way, a linear combination of these shape functions can approximate any sufficiently smooth function on the unit cell with $(N+1)$ th-order accuracy. In this approximation, the function values of the approximated function serve as coefficients of the linear combination, due to the use of Lagrangian shape functions. (Note that this property is pivotal to have both the streaming and the collision step act on the same degrees of freedom.)

The shape functions and support points are transferred from the unit cell to an arbitrary cell D_c via the mapping function M_c , yielding $x_{ci} := M_c^{-1}(\tilde{x}_i)$ and

$$\psi_{ci}(x) := \begin{cases} \tilde{\psi}_i[M_c(x)] & \text{if } x \in D_c, \\ 0 & \text{otherwise,} \end{cases}$$

i.e., defining support points and shape functions on an irregular grid.

3. Reconstruction of the solution

With these preliminaries, the distribution functions can be approximated via

$$f_\alpha(x, t) \approx \sum_{c=1}^{\text{no. cells}} \sum_{i=1}^{(N+1)^{\text{dim}}} \hat{f}_{\alpha ci}(t) \psi_{ci}(x), \quad (3)$$

where $\hat{f}_{\alpha ci}$ are the degrees of freedom, i.e., the values of the distribution function at the (possibly irregular) support points. Inserting this interpolation into the lattice Boltzmann streaming step for $x = x_{ci}$ gives the semi-Lagrangian streaming

$$\hat{f}_{\alpha ci}(t) = \sum_{\zeta=1}^{\text{no. cells}} \sum_{j=1}^{(N+1)^{\text{dim}}} \hat{f}_{\alpha \zeta j}(t - \delta_t) \psi_{\zeta j}(x_{ci} - \delta_t e_\alpha). \quad (4)$$

Note that the sums on the left-hand side have canceled out, due to $\psi_{ci}(x_{cj}) = \delta_{ij}$, rendering the scheme fully explicit.

The right-hand side, i.e., the reconstruction of the solution at the departure point, can be formulated as a matrix vector multiplication,

$$\hat{f}_\alpha(t) = \Psi_\alpha \hat{f}_\alpha(t - \delta_t).$$

4. Algorithm

Our implementation of the SLLBM stores the spatial operators Ψ_α in a sparse matrix. This matrix is assembled upon initialization, which makes up only a small fraction of the total runtime. The assembly routine iterates over all support points of the computational grid. For each support point x_{ci} and particle velocity e_α , it detects the cell D_ζ that contains the respective departure point. A matrix entry $\psi_{\zeta j}(x_{ci} - \delta_t e_\alpha)$ is incorporated into the sparse matrix for each degree of freedom that belongs to the cell D_ζ .

With the matrix assembled, each streaming step is performed by a single sparse matrix vector multiplication $\hat{f}^{(\text{new})} := \Psi \hat{f}^{(\text{old})}$.

D. Stability analysis of the streaming operator

To study the stability of the SLLBM, the spectral radius of the spatial operator was evaluated on a single one-dimensional

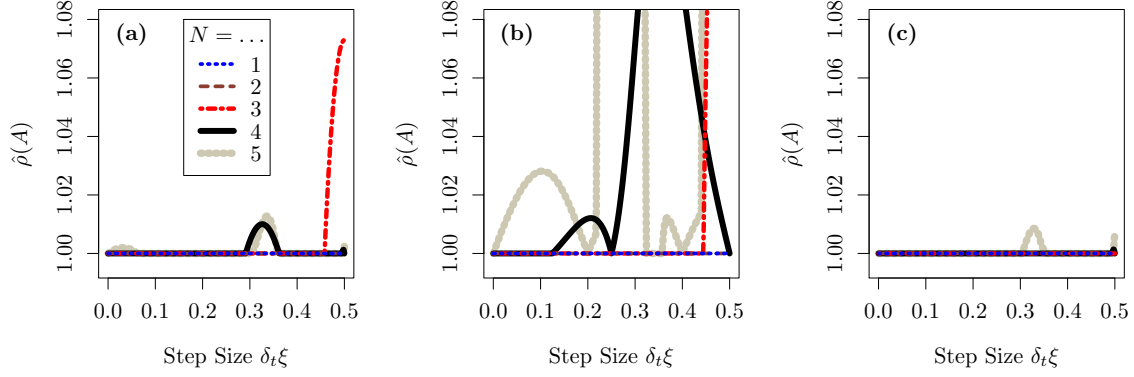


FIG. 2. Spectral radii $\hat{\rho}(A)$ of the spatial operators with respect to the step sizes $\delta_x = \delta_t \xi$. The order of shape functions N and the set of support points were varied. Three sets were considered: (a) Gauss-Lobatto-Legendre, (b) equidistant, and (c) Gauss-Lobatto-Chebyshev points. As spectral radii are symmetric with respect to $\delta_t \xi = 0.5$ and periodic with a period of 1, they are only plotted for $\delta_t \xi \in [0, 0.5]$.

cell. We define the spatial operator as $A(t) = [\tilde{\psi}_j(x_i - \delta_t \xi)]_{ij}$ with periodic boundary conditions $x = x + k$, $k \in \mathbb{Z}$. The streaming step can only be stable if all eigenvalues σ_i of A are contained in the unit circle: $\hat{\rho}(A) := \max_i |\sigma_i| \leq 1$. Note that the exact streaming has a spectral radius of 1.

The spectral radius of A was evaluated for three different sets of support points, namely Gauss-Lobatto-Legendre, equidistant, and Gauss-Lobatto-Chebyshev. The orders of finite elements and step sizes $\delta_t \xi$ were varied. Figure 2 shows that the semi-Lagrangian streaming step is unconditionally stable for $N = 1$ and 2. The stability for higher-order finite elements depends strongly on the set of support points and the step size. The present work uses Gauss-Lobatto-Legendre points, which give a more stable scheme than equidistant support points. However, the Gauss-Lobatto-Chebyshev points enhanced stability even further, entailing unconditional stability even for $N = 3$. This finding indicates that there is still room for improvement in choosing a different set of support points.

An unstable streaming operator means that initial solutions exist, for which iteratively applying the streaming step will drive the solution to infinity. In practice, $\hat{\rho}(A) > 1$ will not necessarily render the simulation unstable, due to the dissipative character of the collision step and the dependence on the initial solution. Furthermore, the use of irregular grids might relax the instability of the scheme. Finally, the present analysis does not predict the stability properties for flows with nonperiodic boundary conditions. Instead, it demonstrates the possibility of an unstable streaming step when using high-order finite elements as well as the unconditional stability for low-order finite elements.

E. Implementation aspects

All simulations were performed using our off-lattice Boltzmann framework NATrium [21]. NATrium builds on the finite-element library deal.II [33] and uses the HPC library Trilinos [22] for the MPI parallelization. On an Intel Xeon E5-2680v3, our D3Q19 implementation of the present SLLBM reaches a performance of 10 million grid point updates per second, i.e., 0.4 million grid point updates per core. When run on a single core, the code reaches a

performance of approximately 1 million grid point updates per second.

III. RESULTS

In the following, the SLLBM is validated for three exemplary flows. We restrict the simulations within this paper to periodic boundary conditions in order to emphasize the high-order convergence properties of the present scheme. The two-dimensional Taylor-Green vortex is utilized to show that the SLLBM solution converges to the incompressible benchmark solution. The doubly periodic shear layers and three-dimensional Taylor-Green vortex are simulated to show the advantage of a higher-order spatial representation in the simulation of intricate flows. The goal of these numerical experiments is to prove that the proposed scheme can simulate complex vortical flows efficiently and accurately.

A. Convergence: Two-dimensional Taylor-Green vortex

The LBM implies numerical errors due to spatial discretization, temporal discretization, and artificial compressibility. As the standard LBM requires equal time steps and grid spacings, the three error terms are coupled. In contrast, the present SLLBM allows the three errors to be studied independently: the spatial error by tuning the computation grid and the order of finite elements, the temporal error by tuning the Courant-Friedrichs-Lewy number $\text{CFL} = \frac{\max |e_\alpha| \delta_t N^2}{\min \delta_x}$, and the compressibility error by tuning the Mach number $\text{Ma} = \frac{U_0}{c_s} = \frac{\sqrt{3}U_0}{\xi}$. Here, δ_x and U_0 are the minimal vertex spacing and characteristic flow velocity, respectively. Note that $\text{CFL} \sim \delta_t$, i.e., large CFL numbers define large time steps. The latter was set to $U_0 := 1$ for all simulations within this work.

The incompressible solution of the two-dimensional Taylor-Green vortex is given as

$$\begin{aligned} u_1^{\text{ref}}(x_1, x_2, t) &= \cos x_1 \sin x_2 \exp(-2\nu t), \\ u_2^{\text{ref}}(x_1, x_2, t) &= -\sin x_1 \cos x_2 \exp(-2\nu t), \\ p^{\text{ref}}(x_1, x_2, t) &= \frac{1}{4}[\sin(2x_1) + \sin(2x_2)] \exp(-4\nu t). \end{aligned}$$

The periodic flow domain $[0, 2\pi]^2$ contains four vortices that decay uniformly due to viscous forces. The viscosity does not

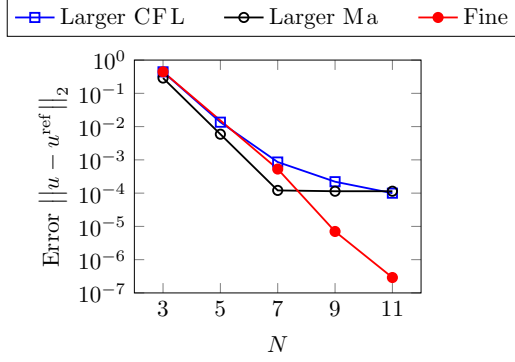


FIG. 3. Convergence of the SLLBM. Errors in simulations of a Taylor-Green vortex with different orders of finite elements N on a 2×2 grid. The “fine” simulation was performed with $\text{Ma} = 0.008$ and $\text{CFL} = 0.1$. The “larger CFL” and “larger Ma” simulations were simulated with $\text{Ma} = 0.008, \text{CFL} = 2$ and $\text{Ma} = 0.1, \text{CFL} = 0.1$, respectively.

influence the features of this particular flow and was thus set to an arbitrary value of $\nu = \pi/5$. The distribution functions were initialized with the equilibrium distributions from $\rho = 1 + p^{\text{ref}}(x_1, x_2, 0)/c_s^2$ and $u^{\text{ref}}(x_1, x_2, 0)$. Errors were measured after the vortices had decayed by one order of magnitude, i.e., $\exp(-2\nu t_{\text{max}}) = 0.1$.

1. Spatial error

To evaluate the spatial accuracy of the scheme, N and δ_x were varied. Figure 3 depicts the decay of the numerical errors with respect to the order of finite elements on a 2×2 grid. Both the temporal error and the compressibility error were kept smaller than the spatial error by setting $\text{CFL} = 0.1$ and $\text{Ma} = 0.008$. With this choice, the SLLBM converged exponentially with respect to the order of finite elements N . At considerably larger CFL or Ma, the temporal error or compressibility error became dominant, respectively.

To supplement the higher-order convergence, simulations for $N = 2$ and 4 were performed at different grid resolutions. Figure 4 shows the deviations from the reference pressure p^{ref} and reference velocity u^{ref} . The slightly scattered convergence rates are due to small-scale oscillations in the solution that are inherent to the LBM. For $N = 2$, the convergence order is already better than that of a standard LBM, i.e., greater than 2. A higher $N = 4$ gives a clear increase in the convergence order.

This result demonstrates that the spatial order of the lattice Boltzmann method can be increased by using a semi-Lagrangian streaming step.

2. Temporal error

The temporal error was evaluated using a 2×2 grid with $N = 13$ and $\text{Ma} = 0.1$, using both the present SLLBM and the previously mentioned SEDG-LBM [19], i.e., a high-order accurate Eulerian OLB. For this high order, the SEDG-LBM enforced small time steps, rendering all simulations beyond $\text{CFL} = 5$ unstable. This behavior is due to the explicit time integration [32], which leads to a CFL constraint of the type $\delta_t \leq C_{\text{CFL}} \max |e_\alpha| (N + 1)^2 / \min \delta_x$, where C_{CFL} is a critical

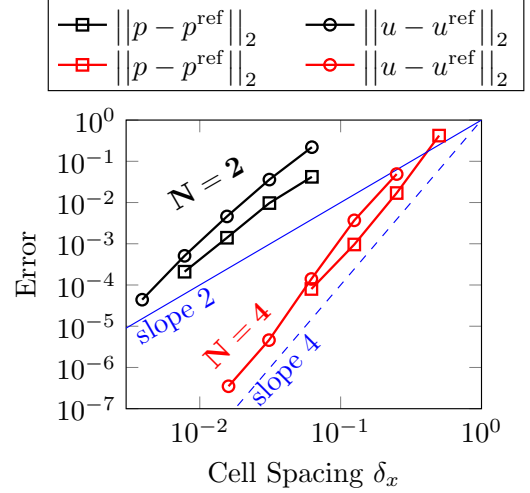


FIG. 4. Convergence of the SLLBM with respect to the grid refinement for $\text{Ma} = 0.008$ and $\text{CFL} = 0.1$. The order of finite elements was $N = 2$ for the black lines and $N = 4$ for the red lines.

CFL number. In contrast, the SLLBM does not incorporate a CFL condition for solving the advection equation. Consequently, it ran stable over a wide range of CFL numbers; cf. Fig. 5. The temporal errors decayed with second order over a range from $\text{CFL} = 100$ to 3.2 . Notably, the simulations were still stable for very large CFL numbers (e.g., 1000, 3000, or 10 000).

3. Compressibility error

Finally, the compressibility error was evaluated at a spatial resolution of 16×16 cells with $N = 7$ and $\text{CFL} = 0.4$, keeping both the temporal and the spatial error small. The Mach number was varied, resulting in the errors that are

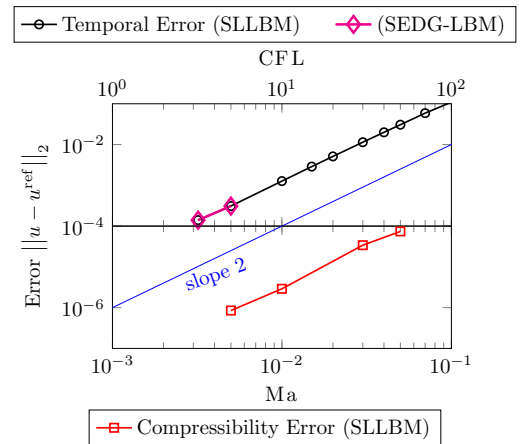


FIG. 5. Top: Convergence of the SLLBM with respect to CFL at constant $\text{Ma} = 0.1$ and spatial resolution (2×2 cells, $N = 13$). For comparison, simulations with the same setup were also performed using the SEDG-LBM [19]. The simulations with the latter were unstable beyond $\text{CFL} = 5$. Bottom: Convergence of the SLLBM with respect to Ma at constant $\text{CFL} = 0.4$ and spatial resolution (16×16 cells, $N = 7$).

depicted in Fig. 5. The compressibility errors converged with $O(\text{Ma}^2)$, as suggested by theory.

B. Vortex evolution: Doubly periodic shear layer at $\text{Re} = 10\,000$

The doubly periodic shear layers were simulated to demonstrate that the SLLBM is able to describe more complex phenomena, such as vortex roll-up, with fewer grid points than previous simulation methods. This benchmark has been the subject of many studies (e.g., [34–37]) as it requires simulation methods to simultaneously capture vortical flow and sharp interfaces. In [35], several different incompressible solvers were applied to the thin shear layer, all of which introduced spurious vortices at a resolution of 128×128 . They could only suppress the evolution of these unphysical vortices for high resolutions of 256×256 grid points or more.

In accordance with their work, the initial flow field in the doubly periodic domain $[0, 1]^2$ was defined as

$$u_1(x_1, x_2) = \begin{cases} \tanh[80(x_2 - 0.25)], & y \leq 0.5, \\ \tanh[80(0.75 - x_2)], & \text{otherwise,} \end{cases}$$

$$u_2(x_1, x_2) = 0.05 \sin[2\pi(x_1 + 0.25)],$$

$$p(x_1, x_2) = 0.$$

The Mach, Reynolds, and CFL number were set to $\text{Ma} = \sqrt{3}/25$, $\text{Re} = 10\,000 = 1/\nu$, and $\text{CFL} = 1$, respectively. Due to the slight perturbation of u_2 , the initial flow became vortical. The simulations were run until $t_{\text{max}} := 1$, i.e., the eddy turnover time. The decay of both kinetic energy and enstrophy were compared to the most accurate simulation in the work by Minion and Brown [35], which was computed with an incompressible pseudospectral method on a 768×768 grid. The kinetic energy and enstrophy were calculated as

$$k = \int_0^1 \int_0^1 \frac{1}{2} u^2 dx_1 dx_2 \quad \text{and} \quad \mathcal{E} = \int_0^1 \int_0^1 (\nabla \times u)^2 dx_1 dx_2,$$

respectively.

First, a grid refinement study was conducted for a low order of finite elements, $N = 2$. Figure 6 shows the decay of kinetic energy and enstrophy for different grid resolutions. The results converged to the reference solution. Figures 7(a) and 7(b) depict the vorticity contours, showing that the spurious vortices disappeared at a resolution of 128×128 cells, i.e., 256×256 grid points.

In a second step, the order of finite elements was enlarged. Simulations were conducted for $N = 11$ on a 16×16 grid, i.e., 176×176 grid points. The vorticity contours of the high-order solution, cf. Fig. 7(c), were free of spurious vortices and resolved more finely than the one in Fig. 7(a). The improved quality of the high-order solution was also reflected by a closer match to the observable properties: Figure 8 shows the decay of kinetic energy and enstrophy in the final phase of the simulation, i.e., $t > 0.8$. While the enstrophy agreed well with the reference solution for all high-order simulations, there was still an offset in the kinetic energy.

To determine the source of this deviation, simulations were conducted with varied spatial resolution, CFL, and Ma . The resulting kinetic energies showed a better match when Ma

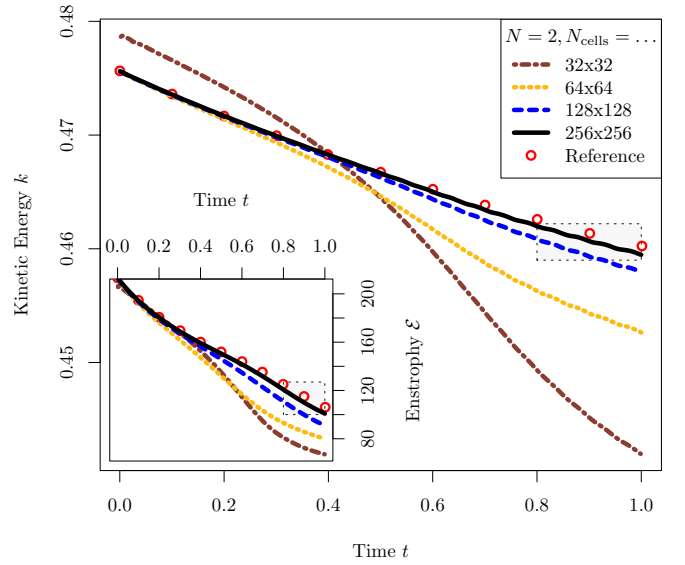


FIG. 6. Decay of kinetic energy k and enstrophy $\mathcal{E}$ in simulations of a doubly periodic shear layer at $\text{Re} = 10\,000$, compared to reference solution [35]. The gray rectangles depict the ranges of Fig. 8, which shows a detailed comparison with the reference in the final phase of the simulations.

was reduced, while varying δ_x or CFL had only a negligible effect. In particular, this result demonstrates the convergence of the temporal and spatial errors, which is supplemented by a comparison to simulation results for the spectral element discontinuous Galerkin LBM.

To emphasize the convergence of the spatial error, the standard deviations of kinetic energy and enstrophy were also compared for different simulation runs; cf. Fig. 9. They are defined as $\text{Std}[u^2]/\langle u^2 \rangle$ and $\text{Std}[\omega^2]/\langle \omega^2 \rangle$ for kinetic energy and enstrophy, respectively. The standard deviations and balances for $N = 2$ converged to the high-order solution. As expected for $N = 2$, the convergence order is 2 for all quantities when the cell spacing δ_x is sufficiently small (here, $\delta_x \leq 0.01$).

In a final simulation, the grid resolution was decreased to 8×8 cells at $N = 13$ and $\text{CFL} = 10$. Figure 7(d) shows that it is possible to obtain good qualitative agreement that is free of spurious vortices. Notably, in spite of the high order, this final simulation took only two minutes on a single core of an Intel Xeon E5-2680v3 and used only 104×104 grid points.

C. Large time steps on an irregular grid: Doubly periodic shear layer at $\text{Re} = 30\,000$

The simulation of the vortex roll-up becomes even more challenging when simulated at a larger Reynolds number. Studies with $\text{Re} = 30\,000$ were performed in [38] and references therein. In the following, we present results obtained with the SLLBM for this higher Reynolds number, and we compare them with results from [38] that were obtained using a standard lattice Boltzmann method with a BGK model on a 256×256 grid.

First, the flow was simulated on regular grids with a high-order discretization ($N = 11$), $\text{CFL} = 10$, and grid sizes of 16×16 and 32×32 cells. Then, simulations with $N = 2$ were

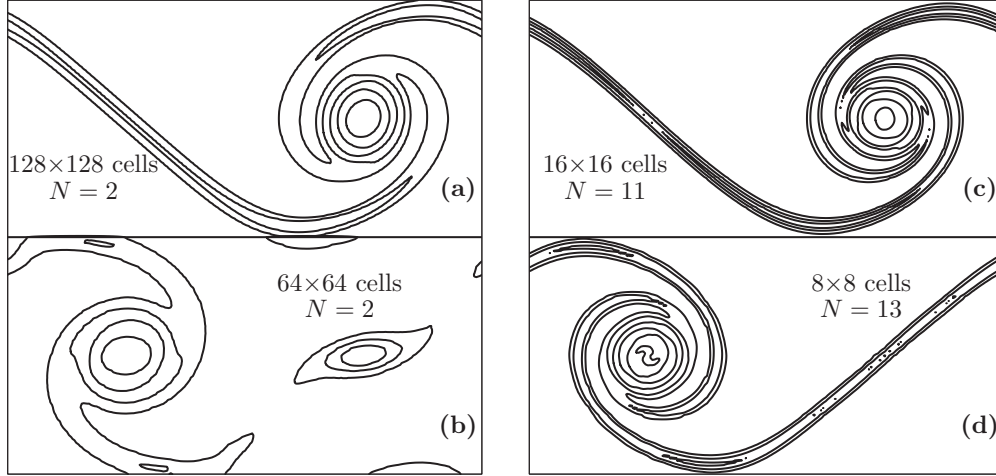


FIG. 7. Vorticity contours of the doubly periodic shear layer at $Re = 10000$ and $t = 1$ from -75 to 75 by an increment of 10 . Due to the symmetry of the flow, the upper plots (a) and (c) show only the upper half of the computational domain, while (b) and (d) show only the lower half. The simulations differed only with respect to the spatial discretization. The numbers of grid points in (a), (b), (c), and (d) were 256×256 , 128×128 , 176×176 , and 104×104 , respectively.

performed on an irregular 256×256 grid for demonstration purposes only. The irregular grid, cf. Fig. 10, was produced using the transformation

$$x = \frac{0.2 \sin(4\pi x_r) + 4\pi x_r}{4\pi},$$

$$y = \frac{0.8 \sin(4\pi y_r) + 4\pi y_r}{4\pi},$$

where (x_r, y_r) denote the nodes of a regular grid. This transformation leads to a smooth refinement in the region of the initial shear interface with a minimum vertex distance of

roughly 7.8×10^{-4} . The CFL number was set to $CFL = 30$, i.e., the time step was more than five times larger than the minimum vertex distance. Such a large CFL number is clearly beyond the stability limit of most, if not all, explicit Eulerian LBMs.

Figure 11 depicts the balances and standard deviations of kinetic energy and enstrophy. The results agreed well with the reference solution. The small offset in the beginning of the simulation might be due to a somewhat different magnitude of the initial perturbation (that is not specified in [38]). While the high-order simulations were slightly less dissipative than the reference results, the low-order simulations on the irregular grid were slightly more dissipative. All in all, the present results closely matched the reference.

D. Transition to turbulence: Three-dimensional Taylor-Green vortex

In contrast to the simple two-dimensional Taylor-Green vortex, the three-dimensional Taylor-Green vortex at $Re = 1600$ develops from single-mode initial conditions into a turbulent flow. As for the previous flows, it has been studied extensively in the literature (e.g., [39,40]). The original work from Brachet *et al.* [39] used a spectral code with 256^3 Fourier modes and is taken as a reference for the SLLBM results. On

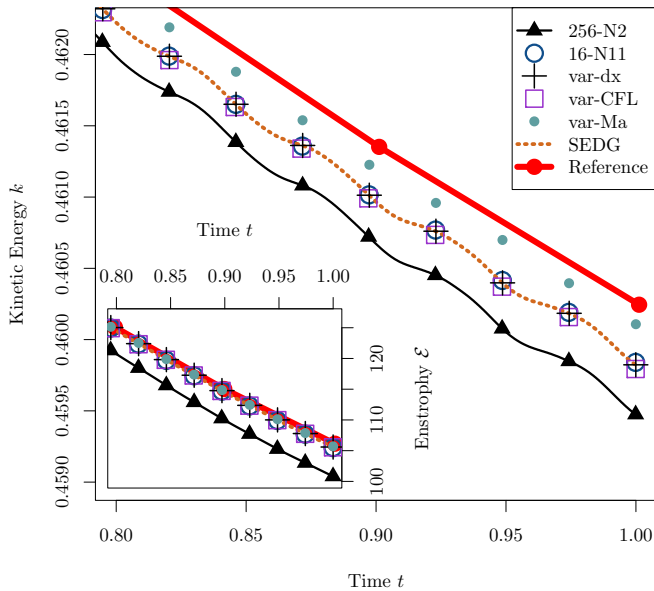


FIG. 8. Decay of kinetic energy k and enstrophy $\mathcal{E}$ in simulations of a shear layer at $Re = 10000$ compared to reference solution [35]. The black triangles (256-N2) depict the most accurate simulation from the previous figure. The other flow parameters for the results shown in this plot are listed in Table I.

TABLE I. Parameters and configurations for shear-layer simulations at $Re = 10000$ that are depicted in Fig. 8.

Name	Method	No. cells	N	CFL	Ma
256-N2	present	256×256	2	1	$\sqrt{3}/25$
16-N11	present	16×16	11	1	$\sqrt{3}/25$
var-dx	present	32×32	11	1	$\sqrt{3}/25$
var-CFL	present	16×16	11	4	$\sqrt{3}/25$
var-Ma	present	16×16	11	1	$\sqrt{3}/37.5$
SEDG	SEDG-LBM [19]	16×16	11	1	$\sqrt{3}/25$
reference	pseudospectral	768×768		adaptive	0

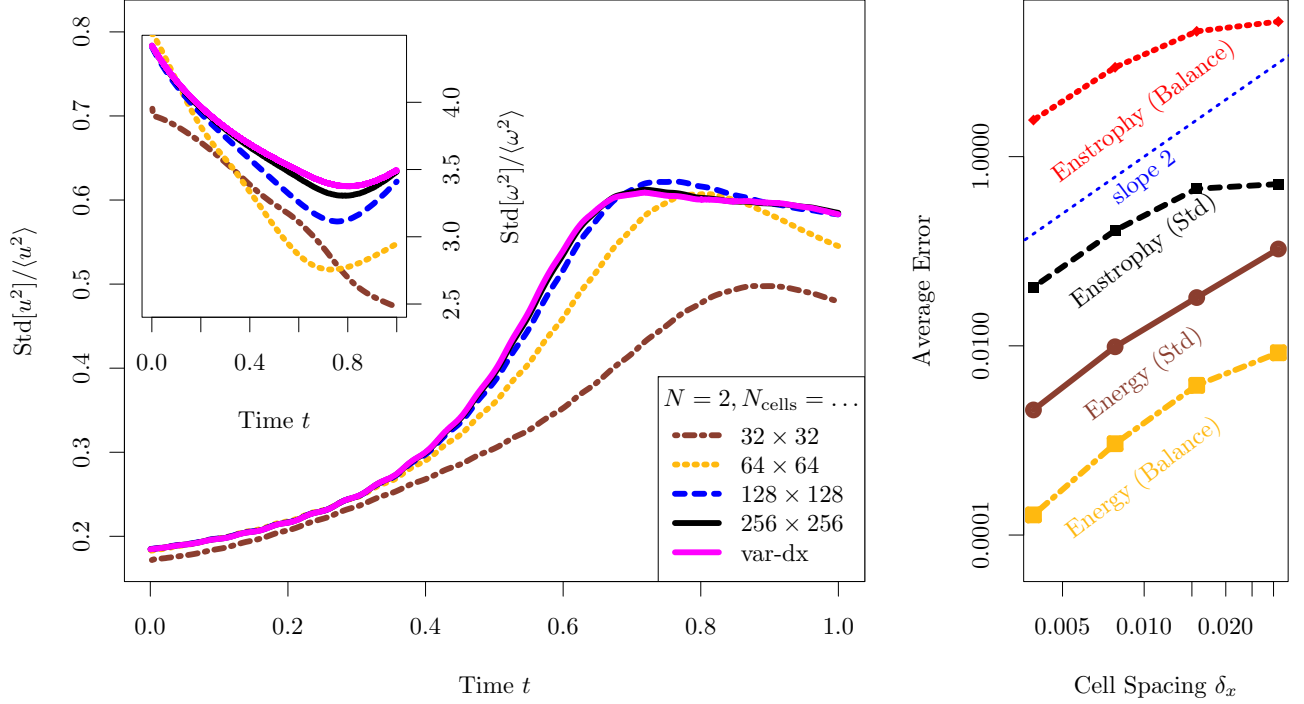


FIG. 9. Left: Convergence of the standard deviations of kinetic energy and enstrophy in simulations of the doubly periodic shear layer at $\text{Re} = 10000$. The flow parameters for the high-order “var-dx” simulation are listed in Table I. Right: L_2 errors of the second-order solutions with respect to the high-order solution “var-dx.” The errors were calculated as $\sqrt{\langle (\Phi - \Phi_{\text{var-dx}})^2 \rangle_t}$, where $\langle \cdot \rangle_t$ denotes a time average and Φ represents the following quantities: k [Energy (Balance)], $\mathcal{E}$ [Enstrophy (Balance)], $\text{Std}[u^2]/\langle u^2 \rangle$ [Energy (Std)], and $\text{Std}[\omega^2]/\langle \omega^2 \rangle$ [Enstrophy (Std)].

the triply periodic domain $[0, 2\pi]^3$, the initial conditions are given as follows:

$$\begin{aligned} u_1(x_1, x_2, x_3) &= \sin x_1 \cos x_2 \cos x_3, \\ u_2(x_1, x_2, x_3) &= -\cos x_1 \sin x_2 \cos x_3, \\ u_3(x_1, x_2, x_3) &= 0, \\ p(x_1, x_2, x_3) &= \frac{1}{16} [\cos(2x_1) + \cos(2x_2)] [\cos(2x_3 + 2)]. \end{aligned}$$

The Reynolds number was defined as $\text{Re} = 1600 = 1/\nu$. The simulations were run with $\text{Ma} = 0.1$ and $\text{CFL} = 2$ on a grid with $32 \times 32 \times 32$ uniform cells. The order of the

shape functions was varied from $N = 2$ to 5, resulting in $160 \times 160 \times 160$ total grid points for $N = 5$.

The turbulent kinetic energy was defined as

$$k = \frac{1}{(2\pi)^3} \int_0^{2\pi} \int_0^{2\pi} \int_0^{2\pi} \frac{1}{2} u^2 dx_1 dx_2 dx_3.$$

Additionally, the turbulent dissipation was calculated as

$$\epsilon = \frac{\nu}{(2\pi)^3} \int_0^{2\pi} \int_0^{2\pi} \int_0^{2\pi} \nabla u : \nabla u dx_1 dx_2 dx_3.$$

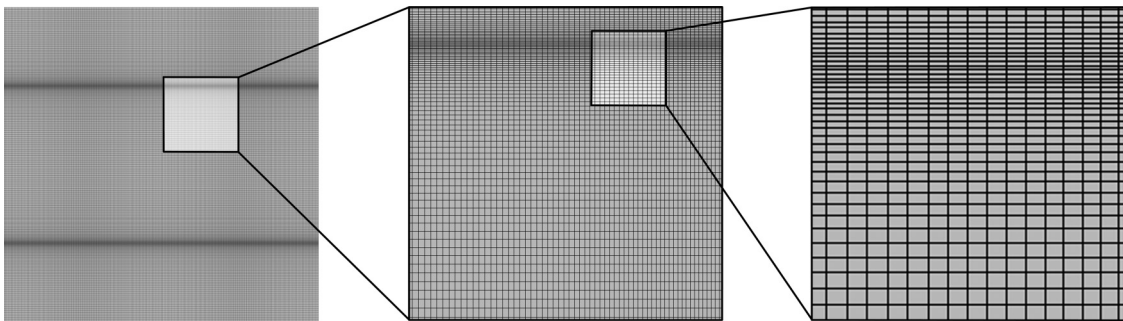


FIG. 10. Irregular computational grid with 256×256 cells. This grid was used in one simulation of the doubly periodic shear layer at $\text{Re} = 30000$ to demonstrate that the SLLBM can be used beyond regular lattices. Note that the general formulation of the SLLBM also allows non-Cartesian grids and grids with hanging nodes to be used.

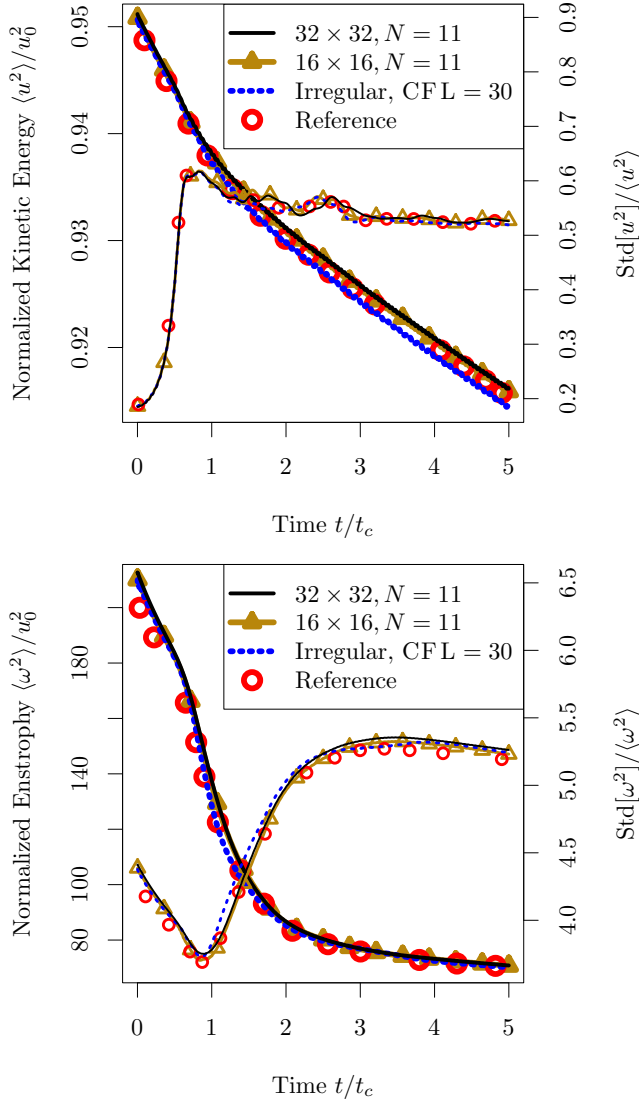


FIG. 11. Normalized kinetic energy and enstrophy in simulations of a doubly periodic shear layer at $\text{Re} = 30\,000$, compared to Ref. [38]. The thick lines and large symbols represent the averages. The thin lines and small symbols represent the standard deviations. The high-order simulations used $N = 11$ and $\text{CFL} = 10$, the “Irregular” simulations used $N = 2$ and $\text{CFL} = 30$ on an irregular grid.

The turbulent kinetic-energy budget is given by $-dk/dt = \epsilon$, as pressure dilatation, turbulent transport, and production vanish in homogeneous flows like this. If the dissipation rate $-dk/dt$ is calculated via finite differences, the left-hand side of the budget equation accounts for all numerical and physical effects that influence the kinetic energy. In contrast, the right-hand side accounts only for the physical dissipation, which is due to viscous effects, as determined via the spatial discretization directly, similar to [41,42]. In line with [40–43], we obtain the dissipation rate $-dk/dt$ via finite differences with an increment of 100 time steps, and we calculate $-dk/dt - \epsilon$ as a measure for the numerical dissipation. Furthermore, to get a slightly different presentation of the

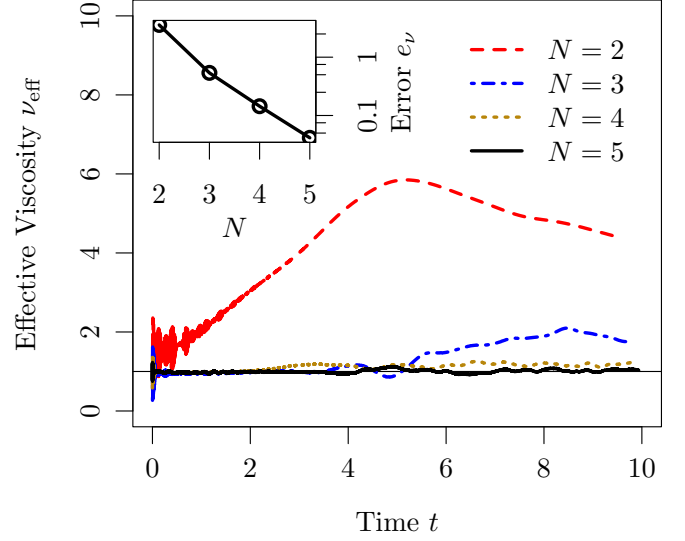


FIG. 12. Refinement study for simulations of a three-dimensional Taylor-Green vortex at $\text{Re} = 1600$. Simulations with $N = 2, 3, 4, 5$ were conducted on a $32 \times 32 \times 32$ grid. The main plot shows the evolution of effective viscosity over time. The error plot in the left upper corner displays the time-averaged error $e_\nu := \sqrt{\langle (\nu_{\text{eff}} - 1)^2 \rangle_t}$ vs the order of finite elements.

results, we define an effective viscosity parameter [41]

$$\nu_{\text{eff}} := -\frac{dk}{dt}/\epsilon$$

that should converge to unity in a fully resolved simulation. In incompressible flows, the turbulent dissipation is also related to enstrophy via [44]

$$\epsilon = \nu \mathcal{E} = \frac{\nu}{(2\pi)^3} \int_0^{2\pi} \int_0^{2\pi} \int_0^{2\pi} (\nabla \times u)^2 dx_1 dx_2 dx_3.$$

This links the effective viscosity parameter to the kinematic viscosity and enstrophy via $\nu_{\text{eff}} = -dk/dt/(\nu \mathcal{E})$.

Figure 12 shows the evolution of the effective viscosity for different orders of finite elements. As the resolution increases, the effective viscosity converged to 1, reflecting that the Navier-Stokes equations were increasingly well recovered at small scales. As for the two-dimensional convergence study, the error decreased in accordance with the order of shape functions.

Figure 13 shows the dissipation of kinetic energy for $N = 5$. For both the SLLBM solution and the reference solution, the kinetic-energy dissipation increased with time, as the spectrum of the flow fills due to smaller and smaller scales being generated. A peak was reached at $t \approx 9$, when the turbulence was fully developed. While the turbulent dissipation was smoother than $-dk/dt$, both agreed closely with the reference. Consequently, the numerical dissipation was small throughout the simulation. The scaled enstrophy agreed almost perfectly with the turbulent dissipation, suggesting that the result was virtually unaffected by artificial compressibility [44]. All in all, the SLLBM captured the growth in kinetic energy accurately and was able to reproduce the point in time and height of the dissipation peak.

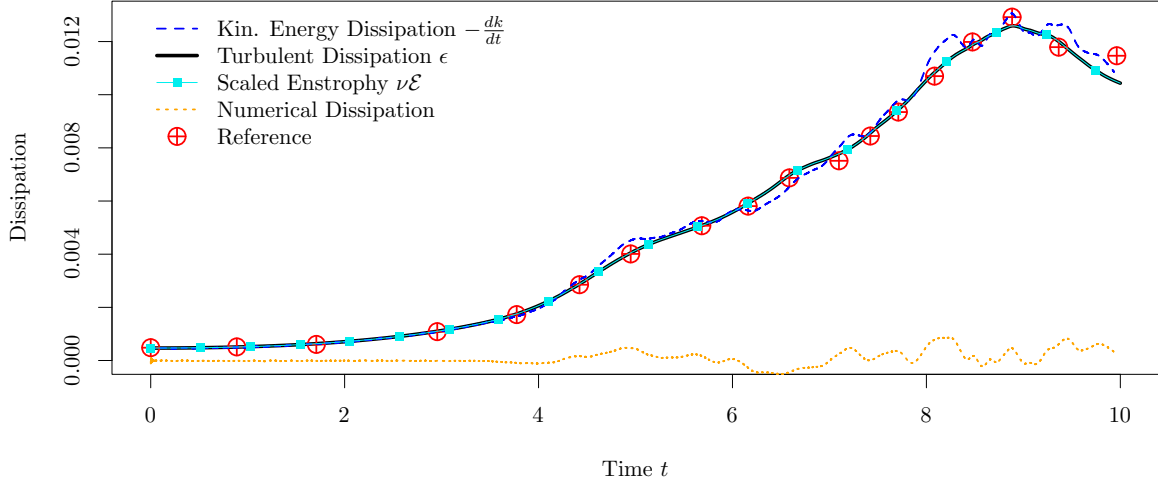


FIG. 13. Kinetic energy dissipation in the simulation of a three-dimensional Taylor-Green vortex at $Re = 1600$. The results of the present SLLBM are compared to the reference solution from Brachet *et al.* [39]. The plot shows the kinetic-energy dissipation $-dk/dt$, the turbulent dissipation ϵ , their difference, the numerical dissipation, and the scaled enstrophy $\nu\mathcal{E}$, which agrees with the turbulent dissipation in incompressible flows.

IV. DISCUSSION

Let us now discuss the quality of our proposed semi-Lagrangian off-lattice Boltzmann method. The present method does not change the collision step but augments the standard streaming step by (a) introducing a higher-order accurate finite-element representation, and (b) decoupling the time step from the computational grid. In the present scheme, the standard LBM appears as a special case for $N = 1$ and $\delta_x = \delta_t$. The SLLBM is unconditionally stable on a regular grid with periodic boundary conditions for $N \in \{1, 2\}$. For larger N , the stability was shown to depend on the choice of the time step size and Mach number. Moreover, switching from Gauß-Lobatto-Legendre to Gauß-Lobatto-Chebyshev support points would further improve the stability properties, yielding unconditional stability for $N = 3$ when using regular grids and periodic boundary conditions.

All but one simulation in this study were performed on regular grids. The goal was to show that the semi-Lagrangian streaming step is a useful generalization of the standard LBM, offering higher-order spatial accuracy and large time steps.

The present scheme was validated for three periodic test cases. Simulations of the two-dimensional Taylor-Green vortex were conducted to study the convergence properties of the present scheme. While the orders of the temporal error and compressibility error accord with the standard LBM [45,46], the spatial errors were demonstrated to converge exponentially with respect to the order of shape functions. The higher-order convergence of the spatial error allows the number of grid points to be reduced while retaining the same accuracy as a standard lattice Boltzmann simulation. This finding conforms to previous higher-order accurate off-lattice Boltzmann methods [18,19,47]. In contrast to the cited methods, the present semi-Lagrangian scheme requires only a single evaluation of the spatial operator per time step, which reduces the computational cost. In addition, the SLLBM does not pose a CFL condition. Theoretically, removing the CFL condition renders the streaming step stable for arbitrarily large

time steps. Practically, this means that the time step is not limited by the advection operator's stability constraint but by the relevant time and length scales of the flow. This is a clear advantage over previous finite-volume or finite-element OLBMs.

These unique benefits of the present scheme were demonstrated in the simulation of doubly periodic shear layers at $Re = 10\,000$ and $30\,000$. These particular flows include sharp features and vortical behavior, making them a challenging problem for fluid simulation codes [35]. For the present scheme with $N = 2$, the kinetic energy and enstrophy converged to the reference solution. As in the original work by Minion and Brown [35], these simulations produced spurious vortices when using fewer than 256×256 grid points. However, when enlarging the order of our scheme to $N = 11$ or 13 , the evolution of unphysical secondary vortices was suppressed even at low resolutions. For example, the final simulation for $Re = 10\,000$ on 104×104 grid points kept the number of grid points roughly six times smaller than previous work [35]. The results for $N = 11$ on a grid with 16×16 cells were in line with the reference solution that was obtained using a pseudospectral code on a 768×768 grid. While previous LBM simulations of the present benchmark [36,38] were only qualitatively compared to the incompressible reference solution, the SLLBM proved to be a close quantitative match.

Furthermore, the simulation results at $Re = 30\,000$ were compared to previous work [38] and closely matched the reference results for this even more challenging setup. One of the simulations ran on an irregular grid, proving that the SLLBM effectively removes the most critical restriction of the standard LBM, i.e., the regular lattice. This particular simulation used a CFL number of 30, which is only possible due to the semi-Lagrangian streaming step. In an Eulerian framework, such a high CFL number would require implicit time integration, which would drastically increase the computational cost per time step [21].

Finally, the SLLBM was used to simulate the transition to turbulence in a three-dimensional Taylor-Green vortex. The evolution of the kinetic-energy dissipation rate agreed closely with a spectral code on a $256 \times 256 \times 256$ grid. A previous study showed that high-order discretizations easily render simulations of this flow unstable, while spatial under-resolution shifts the peak of the dissipation rate [40]. When using a $32 \times 32 \times 32$ grid with $N = 5$, the newly presented SLLBM kept the numerical dissipation small and accurately predicted the dissipation peak, indicating that all relevant scales of the flow were captured.

Taken together, the semi-Lagrangian streaming step presents a promising extension of the LBM, in particular for flows that require high spatial resolutions. A ubiquitous example is the simulation of fluid turbulence. In contrast to the standard LBM, the SLLBM enables unstructured grids to be used, which can be a decisive advantage when it comes to simulating wall-bounded flows or flows in realistic, complex-shaped geometries. The present manuscript lays the

foundation for our further work, which includes applications of the SLLBM to wall-bounded turbulent flows.

V. CONCLUSION

This paper introduces a semi-Lagrangian off-lattice Boltzmann method that decouples the time step from the grid spacing and offers high-order spatial accuracy. The present scheme was validated for challenging two- and three-dimensional flows, illustrating that the semi-Lagrangian streaming step is a valuable alternative to the second-order accurate streaming step that is commonly used in the standard lattice Boltzmann method. A detailed study of the present scheme on irregular grids will be presented in a separate paper. Compared to previous off-lattice Boltzmann methods, the present scheme is computationally efficient, which will be shown in detail in a subsequent publication [21]. Furthermore, the semi-Lagrangian streaming step can be combined with other collision models, which will also be part of our future research.

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